

DNS Protocol

What is DNS?

The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6)

How DNS Works Step by Step:

- 1- User Request:** When you type a domain name (for example, `www.example.com`) into your browser, your device begins the process of looking up its IP address.
- 2- Cache Check:** The DNS resolver first checks its own cache (or the system or network cache) for the IP address. If it finds it, it returns the address immediately.
- 3- Query to Root Server:** If the cache does not have the address, the resolver sends the request to a root server. The root server does not know the IP address but directs the request to the top-level domain (TLD) server responsible for ".com," ".org," or ".net."
- 4- Query to Top-Level Domain (TLD) Server:** The resolver forwards the request to the TLD server for the requested domain (such as ".com"). The TLD server identifies the authoritative server responsible for the subdomain (like `example.com`) and directs the resolver to it.
- 5- Query to Authoritative Server:** The resolver finally sends the request to the authoritative name server for `example.com`. This server holds the DNS records (such as the A record) that link the domain name to the specific IP address.
- 6- Returning the IP Address to the Browser:** The authoritative server replies with the IP address for `www.example.com` to the resolver, which then passes it to the browser.
- 7- Connecting to the Server:** The browser uses this IP address to connect to the web server and load the webpage for you.

Types of DNS Servers:

1. Root Server

Function: It is the apex of the DNS hierarchy, directing initial queries to the correct TLD (Top-Level Domain) servers based on the domain extension (such as .com, .org).

Importance: The starting point for all DNS lookups worldwide, with only 13 root server clusters existing globally.

2. Top-Level Domain (TLD) Server

Function: Manages all domains under a specific extension (such as .com, .net, .org, .sa). It directs the request to the appropriate authoritative name server for the subdomain (such as `google.com`).

Importance: Constitutes the second layer in the DNS hierarchy and helps identify the authoritative server for the requested domain.

3. Authoritative DNS Server

Function: Contains the definitive and final records for a specific domain (such as `google.com`) and provides the final answer—typically the IP address—for the query.

Importance: Serves as the ultimate source of information for the domain, completing the resolution process.

4. Recursive Resolver (DNS Resolver)

Function: Receives the user's request and initiates the full lookup process (querying the root, then the TLD, then the authoritative server), then returns the result to the user. It acts as an intermediary.

Importance: Simplifies the process for users by managing the sequence of queries, often operated by an Internet Service Provider (ISP).

How do they work together? (Example: looking up google.com)

DNS Records

A Record (Address Record):

To translate a domain name (such as example.com) into a numerical IP address (IPv4) of the server hosting the website. It is the primary record linking the domain to the server.

AAAA Record (Quad-A Record):

Similar to the A record, but it links the domain name to an IPv6 address (the next-generation IP address format).

CNAME Record (Canonical Name Record):

To create an alias for another domain. It is used to direct a subdomain (such as blog.example.com) to another main domain (such as example.com) instead of an IP address.

MX Record (Mail Exchanger Record):

Specifies the mail server responsible for receiving email messages for the domain. It can assign priority to multiple mail servers.

NS Record (Name Server Record):

Identifies the trusted name servers that hold the DNS records for the domain. These servers are responsible for providing domain information.

DNS Security

DNS is a core component of the Internet, translating domain names into IP addresses, and securing it is essential to protect communications.

DNSSEC adds digital signatures to DNS records, ensuring their authenticity and preventing tampering such as DNS spoofing or cache poisoning. Additionally, DNS over HTTPS (DoH) and DNS over TLS (DoT) encrypt user queries, enhancing privacy and security, especially on public networks.

With these mechanisms, users and organizations can rely on DNS as a secure and trustworthy way to route communications across the Internet.

Attacks on DNS

Despite its critical role in the Internet, the Domain Name System (DNS) is a common target for cyberattacks.

These attacks exploit weaknesses in DNS to disrupt services or redirect users to incorrect destinations.

One of the most common attacks is **DNS spoofing**, where attackers send fake DNS responses to users or servers, redirecting them to malicious or fraudulent websites.

Another serious attack is **cache poisoning**, in which false DNS records are injected into a resolver's cache, affecting many users for an extended period.

In addition, **Denial of Service (DoS)** and Distributed Denial of Service (DDoS) attacks aim to overwhelm DNS servers with massive amounts of traffic, causing websites and online services to become unavailable. Other attacks, such as DNS amplification, further increase the impact of these attacks by abusing DNS responses.

Real-world Examples

Real-world DNS attacks highlight the serious risks these threats pose to both users and organizations.

1- Dyn DNS Attack (2016): A botnet targeted a DNS provider, causing major websites such as Twitter, Netflix, and GitHub to become temporarily unavailable, affecting millions of users worldwide.

2-Cache Poisoning Attacks: For example, the vulnerability discovered by Dan Kaminsky (2008) allowed attackers to inject false DNS records, redirecting users to fraudulent websites and compromising sensitive data.

3- Root DNS Servers Attack (2020): Attackers attempted to disrupt some of the root DNS servers, affecting response times for some local ISPs and highlighting the need to secure the critical DNS infrastructure against future attacks.

These examples demonstrate that DNS attacks can disrupt services, cause financial losses, damage organizational reputation, and directly impact everyday users.

Conclusion

The Domain Name System (DNS) is a fundamental component of the Internet, facilitating the translation of domain names into IP addresses and ensuring seamless access to services. Studying how DNS works, its different types, and records highlights its complexity and critical role in network operations.

However, DNS is vulnerable to various attacks such as DNS spoofing, cache poisoning, and DDoS, emphasizing the importance of security mechanisms like DNSSEC and encryption techniques such as DoH and DoT. Real-world attack examples demonstrate their significant impact on users and organizations, including service disruptions and data or financial losses.

Understanding DNS—from its structure and operation to potential threats and protective measures—is essential for network and cybersecurity professionals, as well as organizations and users aiming to maintain reliable and secure Internet communication.